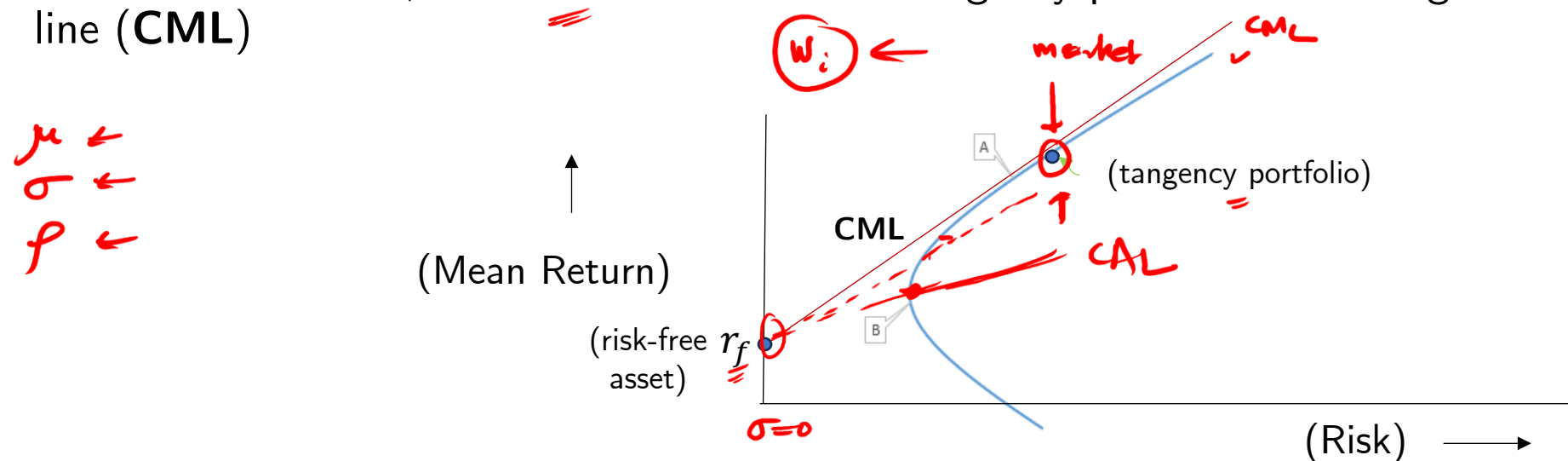


Portfolio Management (2. Factor Models)

Capital Market Line



- ▶ On the efficient frontier, tangency portfolio is a purely risky portfolio with specific proportions of risky assets. It's the best risky portfolio because it has the highest Sharpe ratio
- ▶ Assuming all investors have homogeneous expectations of return, risk, correlations and have the same investment horizon etc., everyone would want to include the tangency portfolio and combine it with risk-free asset.
- ▶ If everyone identifies and buys the same tangency portfolio, then in equilibrium, the relative market values of the assets will adjust to reflect their allocation
- ▶ In these conditions, the market becomes the tangency portfolio. The tangent is called Capital Market line (**CML**)



Computational Issues with Mean Variance Analysis

► One practical problem with mean variance analysis is the dimensionality problem.

► If we have N securities, we need to estimate

- N expected returns μ
- N standard deviations σ
- $N(N-1)/2$ correlations

μ
 σ
 ρ

N

$$N_{C_2} = N(N-1)/2$$

N^2

The number of parameters blow up when N gets larger $\mathcal{O}(N^2)$

► One classic approach to reduce dimension is factor models. (e.g., PCA). In Factor models, asset returns are decomposed into a parsimonious set of factor(s) returns.

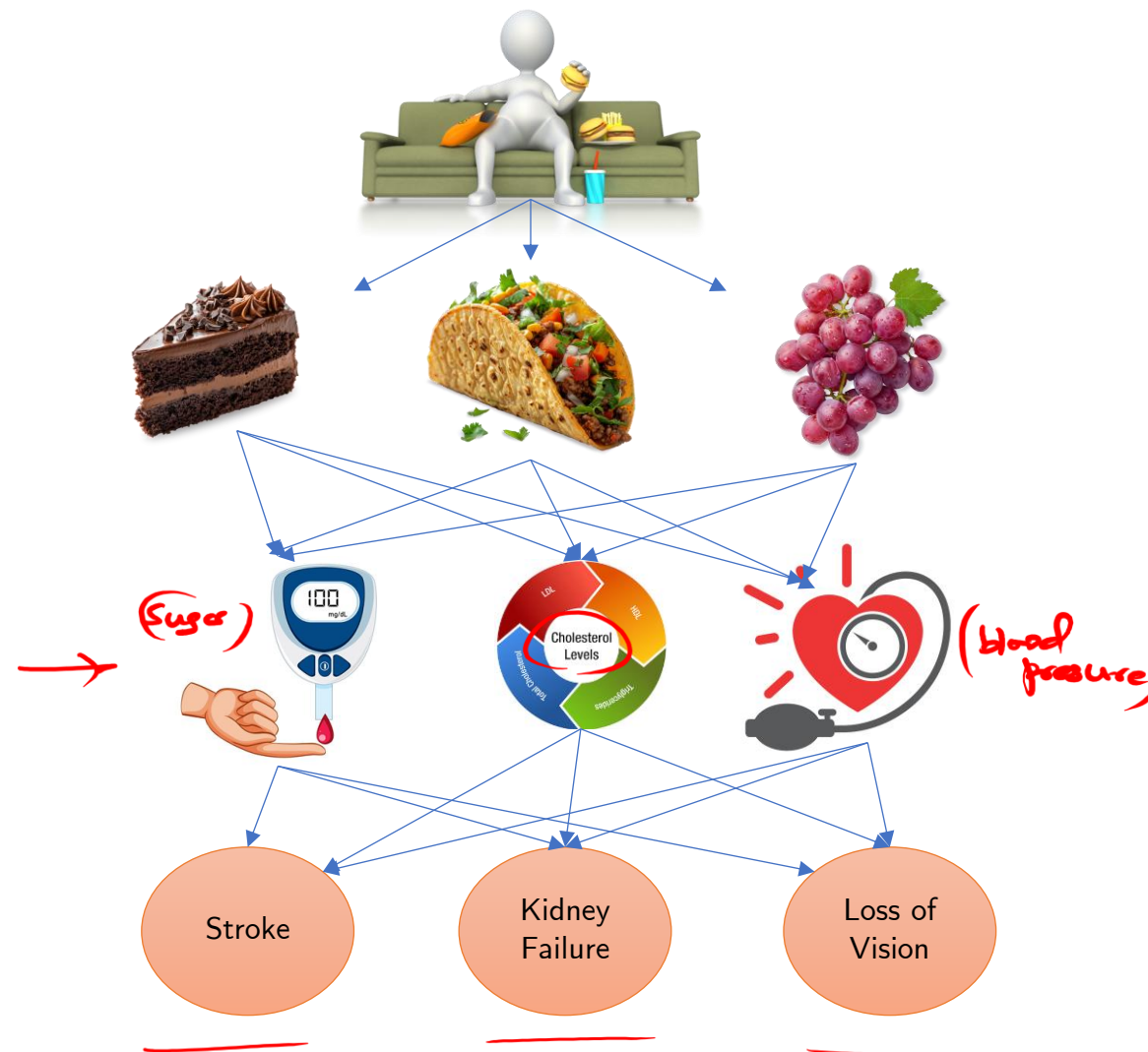
A striking analogy

(Consumer)

(Food)

(Vitals)

(Impacts)



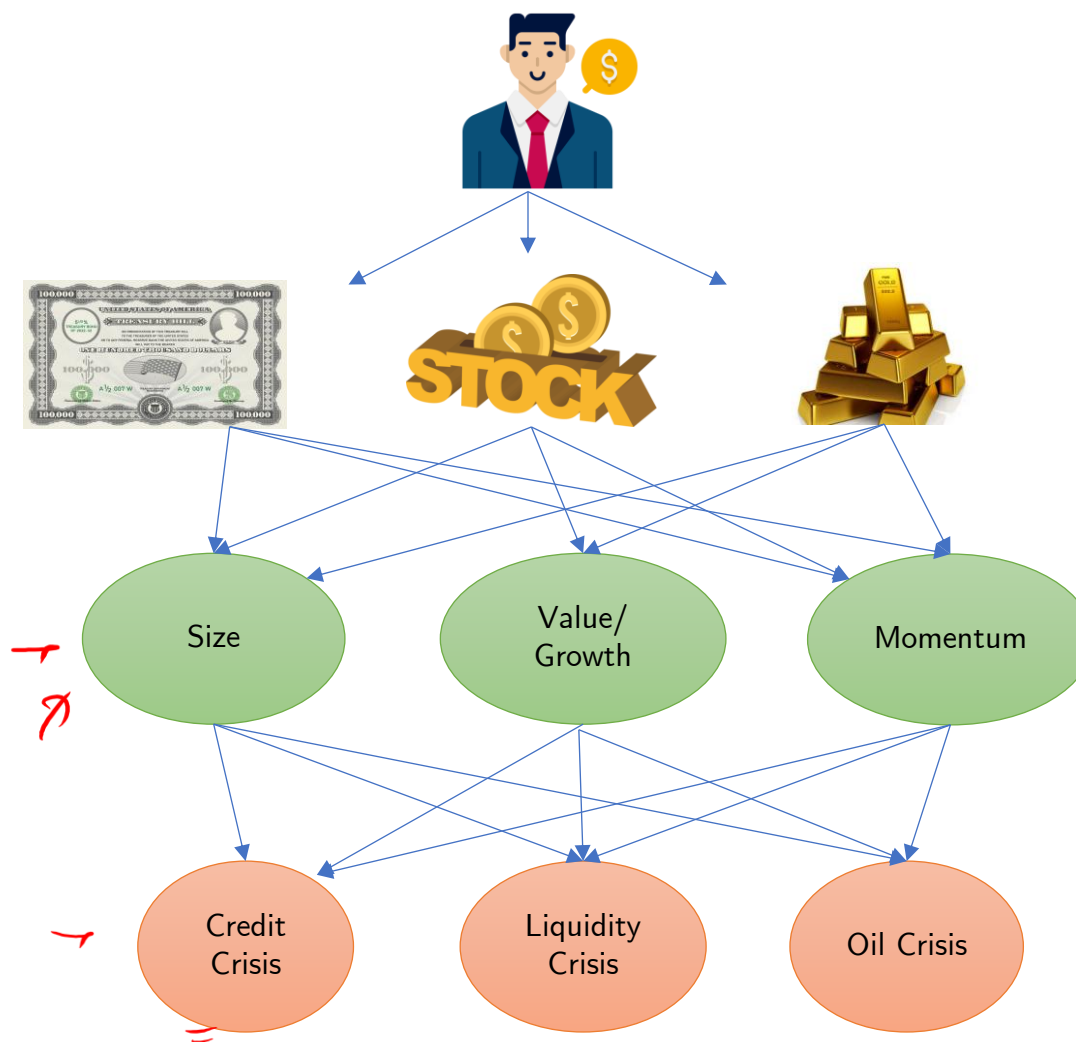
A striking analogy

(Investor) ✓

(Assets) ✓

(Factors) ✓

(Crisis) ✓



Factor Theory



- ▶ Factors matter, not the Assets. Assets are just bundles of factors
- ▶ Factors are flavors of bad times
- ▶ Investors get risk premium for bearing losses during the bad times
- ▶ Different investors have different optimal exposures to set of risk factors

An ad-hoc linear factor model

- ▶ A generic linear factor model is expressed as a simple regression model

$$R_i = \alpha_i + \beta_i(R_m) + \epsilon_i$$

(Expectation)

$$\mathbb{E}[R_i] = \alpha_i + \beta_i(\mathbb{E}[R_m]) \quad \mathbb{E}[\epsilon_i] = 0$$

(Variance)

$$\mathbb{V}[R_i] = \beta_i^2(\mathbb{V}[R_m]) + \mathbb{V}[\epsilon_i]$$

(Total Risk) (Systematic Risk) (Idiosyncratic Risk)

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)} = \rho_{i,m} \left(\frac{\sigma_i}{\sigma_m} \right)$$

- ▶ The ad-hoc one factor model does not make much of an economic sense.

Question



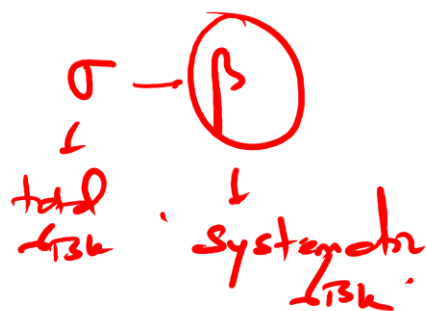
Exercise 1 - In an ad-hoc single factor model

- What is the correlation 2 asset returns ?
- What is the total number of parameters to be estimated in a portfolio of N assets ?
- What is the expectation and variance of a portfolio of N assets ?
- What happens when $N \rightarrow \infty$?

CAPM



- ▶ Capital Asset Pricing Model (CAPM) developed by William Sharpe, Jack Treynor. Sharpe and Markowitz won Nobel Prize in 1990.
- ▶ CAPM is a linear factor model, where 'market premium' is taken as the single factor.
- ▶ Before CAPM the risk of an Asset was viewed in isolation i.e., volatility or variance of the asset. But CAPM introduces the notion of risk as a relative measure. The risk of an Asset in CAPM is the exposure to the market factor. In other words, how the asset return move in relation to the market return.
- ▶ CAPM's main idea is that Asset risk premiums are driven by factor risk premiums that the asset is exposed to. Although empirically CAPM is a failure, it's still a workhorse model for many asset management companies today with a few extensions (e.g., multi-factor model)



active return

$$\underbrace{\mathbb{E}[R_i] - R_f}_{\text{(Asset Risk premium)}} = \underbrace{\beta_i}_{\text{food}} \underbrace{(\mathbb{E}[R_m] - R_f)}_{\text{(Market Risk premium)}} \quad \text{(CAPM)}$$

CAPM vs ad-hoc linear model

- ▶ CAPM has two differences when compared with an ad-hoc regression model.
- ❑ The returns are measured in excess of risk-free return, which indicates a premium. And the premium comes because of exposure to market premium. So basically, you get proportionally rewarded for each unit of active risk.
- ❑ $\alpha = 0$, which means no abnormal return beyond the compensation received for taking systematic risk. It makes sense, as one can fully diversify idiosyncratic risk, so no credit for taking idiosyncratic risk.
- ▶ In practice, the CAPM model is fitted on return data as

$$R_i = \alpha_i + \beta_i R_m + \epsilon_i$$

$$R_i - R_f = \beta_i (R_m - R_f) + \epsilon_i$$

active active

$$E[R_i] - R_f = \beta_i (E[R_m] - R_f)$$

Lessons from CAPM

- ▶ Hold the factor not the Asset (It's only the exposure to the market factor that drives the risk premium, not the idiosyncratic component)
- ▶ CAPM result can be directly derived from mean-variance analysis. Investors lie on the CML by holding the tangency portfolio and the risk-free asset in different proportions based on their risk-aversion
- ▶ An average investor holds the market portfolio. The market risk premium is tied to the variance through the average risk-aversion parameter $\bar{\gamma}$



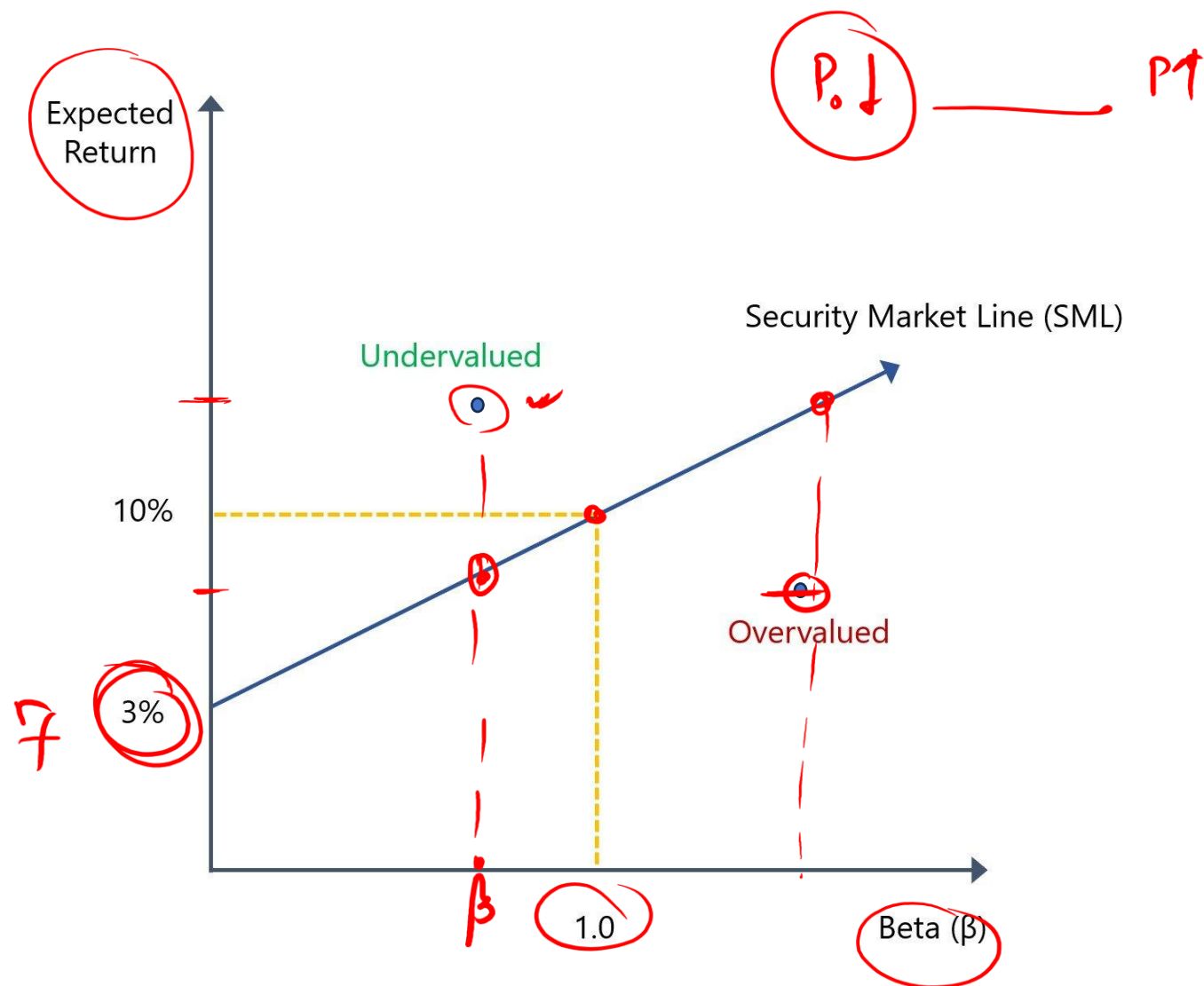
$$\bar{\gamma} =$$

$$\mathbb{E}[R_m] - R_f = \bar{\gamma} \sigma_m^2$$

- ▶ Risk is factor exposure (not the variance of the asset). The only risk that's relevant is the systematic risk. The asset premium is driven by exposure to this systematic risk. Higher the exposure, higher the expected return. This relation is depicted by Security Market Line (SML)
- ▶ Assets that pay off at bad times have low risk premiums. These assets are attractive (gold)

$$\bullet \beta < 0 \quad \sigma \rightarrow \beta$$

Security Market Line (SML)



$$\underline{E[R_i]} = \underline{R_f} + \underline{\beta_i} (\underline{E[R_m]} - \underline{R_f})$$

Failures of CAPM

- ▶ Investors have only financial wealth (e.g., human capital, labor income is ignored)
- ▶ Investors have mean-variance utility (e.g., asymmetric utility functions are more realistic)
- ▶ Investors have a single period investment horizon (With multi-period horizon, other optimization techniques such as stochastic optimization will be more appropriate that considers rebalancing)
- ▶ Investors have homogeneous expectations (in absence of this the market portfolio won't be mean-variance efficient)
- ▶ No taxes or transaction costs
- ▶ Individual Investors are price takers (prices are driven by market demand and supply, one individual cannot move the market price)
- ▶ Information is cost-less

MVE

$$U(\mu, \sigma) = \mu - \frac{1}{2} r \sigma^2$$